

PLATFORM INTELLIGENCE BRIEF — CONFIDENTIAL

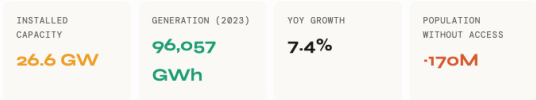
GridIQ West Africa

AI-powered grid loss intelligence and real-time energy data infrastructure for ECOWAS distribution utilities.

- ECOWAS
- IoT + AI
- NTL Detection
- Energy Access
- DFI-Ready
- RenewBiz · AFCEN · Smart Innovation

01 The Opportunity

West Africa sits at an inflection point in its energy transition. Generation capacity is growing — yet the infrastructure to convert that capacity into commercial revenue remains broken. The opportunity is not to build more power; it is to capture the value from what is already being generated.



Despite a 57.7% electrification rate across a population of 400 million, the region's utilities are trapped in a structural paradox: the more electricity they produce, the larger their unrecovered losses become. GridIQ breaks this cycle by transforming invisible loss into measurable, recoverable revenue — and by turning siloed utility data into a live, credible signal for DFIs, regulators, and investors.

The core market thesis: Revenue recovery from existing generation is faster, cheaper, and less risky than adding new generation capacity. A utility recovering just 5 percentage points of NTL recaptures millions of dollars annually — without a single new megawatt installed.

02 The Problem Being Solved

Two crises compound each other across the ECOWAS grid: runaway non-technical losses and near-total energy data blindness. Neither can be solved without the other.

Grid Loss Crisis

The ECOWAS regional average for transmission and distribution losses stands at 21.3% — more than double the international benchmark of 8–10%. For every 10 units generated, more than 2 are lost, stolen, or unaccounted for before reaching a paying customer.

UTILITY	TOTAL LOSSES	NTL BAR	STATUS
EDQ (Guinea)	41.0%	31.9%	Critical
EDSA (Sierra Leone)	~38.0%	~25%	Critical
ECG (Ghana)	~28.4%	~17%	High
NEDCO (Ghana)	~26.5%	~14%	High
NAWEC (Gambia)	~22.7%	~10%	High
SENELEC (Senegal)	~17.4%	~5%	Moderate
CIE (Cote d'Ivoire)	8.7%	~3%	Best Practice

The ECOWAS EIS Data Blindness Gap

Even where losses are partially understood at utility level, this knowledge does not reach ECOWAS regulators, DFIs, or national planning ministries. The ECOWAS Energy Information System relies on voluntary, manual reporting, with lags of 2–8 years across most member states. DFIs conducting due diligence cannot obtain current, verified utility performance data — forcing higher risk premiums and, in some cases, declining to invest altogether.

The Causal Chain

No feeder-level metering	Losses invisible — no geographic attribution possible
Losses undetected	No targeted enforcement; NTL normalises and scales
Revenue shortfall	Non-cost-reflective tariffs + unrecovered NTL = insolvency risk
Utility debt builds	Cannot service PPA debt; cannot invest in grid maintenance
Grid reliability falls	More self-generation on diesel; utility revenue base shrinks
Private capital exits	DFI lending constrained; energy access stagnates

03 Platform Architecture — The Full Stack

GridIQ is a cloud-hosted, IoT-enabled AI platform operating across three integrated functions — Detect, Recover, Report — delivered through six technical components that together form the region's first purpose-built grid intelligence stack.

Smart Meter Analytics

AI/ML analysis of Conlog/Hexing prepaid token transaction histories. Flags billing anomalies, meter

Supervised ML

Engine	bypassing, and theft — calibrated for token-purchase event streams, not monthly billing cycles.	
IoT Grid Node Monitoring	10 ruggedised smart sensors at distribution transformers and primary feeders. Captures voltage, current, active/reactive power, and power factor at 15-minute intervals with 48-hour battery backup.	4G / LoRaWAN
Real-Time GIS Dashboard	Six-module responsive web app — live loss map, zone drill-down, alert feed, investigation queue, monthly PDF reports, and one-click EIS export. English/French, 2G/3G-optimised.	Responsive Web
Automated EIS Reporting Pipeline	Structured monthly data feed from each utility's GridIQ instance to the ECOWAS EIS via REST API — replacing manual, lagged, and frequently absent reporting with a live, co-validated pipeline.	REST API / JSON
Predictive Maintenance Alerts	ML anomaly detection on continuous sensor data identifies transformers showing early degradation — power factor below 0.85, voltage deviation outside ±5%, load factor above 80%, asymmetric phase loading.	ML Anomaly Detection
Impact ROI Tracker	Transparent, auditable revenue recovery accounting. Exportable to PDF and Excel in AFD, AIDB, IFC, and EU reporting formats. Makes grant accountability automatic rather than retrospective.	AFD/EU Formats

04 The AI Opportunity Engine — Core Differentiator ▾

GridIQ's competitive core is its AI intelligence layer — the set of machine learning capabilities that transform raw sensor and meter data into actionable commercial intelligence. Four distinct AI functions work in concert.

 Token-stream NTL detection The only NTL engine in the region calibrated for prepaid token-purchase event streams — the dominant metering paradigm across ECOWAS. Detects meter bypass, illegal reconnection, and billing manipulation without requiring monthly cycle data.	 Georeferenced loss attribution Compares injected power against metered consumption at feeder and transformer level. AI engine isolates loss zones in real time and generates georeferenced alerts identifying specific clusters for field investigation.
 Predictive maintenance ranking Continuous monitoring of voltage deviation, power factor, load factor, and phase asymmetry across all instrumented transformers. Monthly maintenance priority rankings sent to engineering teams — estimated to reduce unplanned outages by 15–25% by Year 2.	 Revenue recovery quantification Real-time calculation of recovered revenue attributable to platform interventions: (Baseline NTL kWh – Current NTL kWh) × utility tariff. Cumulative monthly tracker with DFI-formatted export — creating an auditable accountability record for donors and lenders.

What makes this hard to replicate: The AI models are trained on the specific token-purchase event structures of Conlog and Hexing prepaid systems — the two dominant meter vendors in the ECOWAS market. Adapting generic anomaly detection to this paradigm requires proprietary training data and deep utility integration that cannot be acquired quickly.

05 Revenue Model — All Monetisation Streams ▾

GridIQ is structured to generate revenue across multiple streams, combining platform subscriptions with performance-linked recovery fees and grant-funded deployment tranches — creating a resilient, multi-layer commercial model suited to both public utility clients and development finance cycles.

Utility SaaS Subscription Annual platform licence per utility — covering dashboard access, AI analytics, EIS reporting pipeline, and platform maintenance. Scales with feeder count and zone coverage.	SaaS / Annual
IoT Hardware Deployment One-time or leased provision of ruggedised smart sensors, LoRaWAN gateways, and installation services. Optionally bundled with platform licence in a capex-lite model for constrained utilities.	Hardware / Lease
Revenue Recovery Share Performance-based fee tied to verified NTL reduction. A percentage of auditable recovered revenue flows to GridIQ — aligning platform incentives directly with utility commercial outcomes.	Performance %
Grant & DFI Project Fees Implementation and technical assistance fees embedded in AIDB, EU, AFD, and IFC-funded energy access programmes. The ROI Tracker's DFI-format reporting makes GridIQ a natural fit for grant-funded pilots that require impact accountability.	Grant / TA Fees
ECOWAS EIS Data Services Aggregated, anonymised regional grid performance data offered as a subscription intelligence feed to ECOWAS regulators, energy ministries, researchers, and private investors — with GridIQ as the region's primary real-time EIS provider.	Data Licence
Maintenance & Managed Services Ongoing sensor calibration, firmware updates, field support contracts, and quarterly performance reviews. Particularly relevant for utilities with limited in-house technical capacity.	Recurring Services

06 Platform & Product Features ▾

The GridIQ dashboard delivers six integrated modules, each purpose-built for the operational realities of ECOWAS utilities — designed for 2G/3G connectivity, bilingual interfaces, and field-first workflows.

F-01 Live Loss Map Feeders colour-coded by NTL severity, refreshed every 15 minutes. Geographic view of the entire pilot zone.	F-02 Zone Drill-Down Injected vs. billed energy per zone with trend charts. Enables root-cause analysis at feeder and transformer level.	F-03 Alert Feed Real-time theft signatures and tamper events surfaced as prioritised alerts — with time-stamp, location, and account ID.
F-04 Investigation Queue Ranked customer account list with GPS coordinates, ready to assign to field teams for physical verification and enforcement.	F-05 Monthly PDF Reports Auto-generated performance reports for utility management — covering NTL trends, revenue recovery, and maintenance priorities.	F-06 EIS One-Click Export Structured submission to the ECOWAS EIS in standardised JSON schema. Utility co-validates before each submission.
F-07 Maintenance Rankings Monthly priority list of transformers showing degradation signals — delivered to engineering teams for pre-failure intervention.	F-08 Impact ROI Tracker Live cumulative revenue recovery accounting in AfDB, IFC, AFD, and EU export formats — making grant reporting automatic.	F-09 Bilingual Interface Full English/French interface. 2G/3G-optimised rendering for use in network-constrained utility environments across the region.

07 Competitive Landscape & Moat

GridIQ competes in a market where the main alternatives are expensive Western smart grid platforms built for fully-metered, digitised utilities — a profile that describes almost no ECOWAS utility today. Generic platforms require deep AMI infrastructure that does not exist; they are also not built for the token-purchase metering paradigm. GridIQ’s moat is structural, not just technological.

Prepaid token-stream AI The only NTL detection engine adapted to Conlog/Hexing token-purchase event streams — the dominant meter paradigm across ECOWAS. Competitors require monthly AMI data that most utilities do not have.
ECOWAS EIS integration GridIQ is the only platform with a direct, standardised API connection to the ECOWAS Energy Information System — making it the path of least resistance for any ECOWAS member utility seeking to meet reporting obligations.
Context-specific hardware stack IP65 ruggedised sensors with LoRaWAN fallback and 48-hour battery backup are designed specifically for West African substation environments — not repurposed from European or North American deployments.
DFI reporting as a native feature The Impact ROI Tracker generates grant accountability reports in AfDB, IFC, AFD, and EU formats as a built-in platform function — a capability no other grid analytics platform has engineered for this market.
Low-bandwidth, bilingual design The platform is 2G/3G-optimised with a full English/French interface — two design constraints that eliminate all major Western competitors from realistic deployment consideration.

08 Go-to-Market Strategy

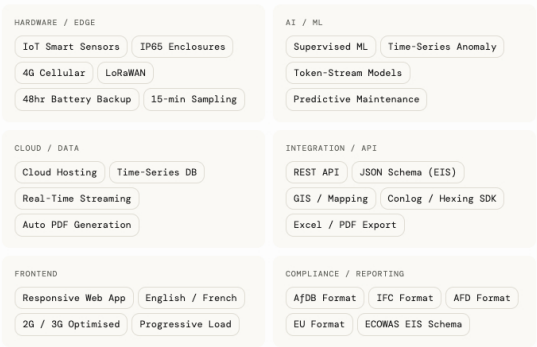
GridIQ enters the market through a pilot-anchor strategy, using a single high-visibility utility deployment to generate verified loss reduction data, DFI relationships, and regional credibility — then scaling across the ECOWAS corridor on the strength of audited outcomes.

Phase 1 · Yr 1	Pilot Deployment — EDSA (Sierra Leone) Single utility pilot zone. 10 IoT sensors deployed. Baseline NTL measurement established. AI model calibrated to utility’s Hexing/Conlog token data. First live EIS report generated. Recovery ROI documented for DFI reporting.
Phase 2 · Yr 2	Expansion — ECG & NEDCO (Ghana) Leverage Sierra Leone outcomes data to onboard Ghana’s two distribution utilities. Expand sensor network. Refine AI models on a second national metering dataset. Scale DFI grant pipeline through AfDB and EU Energy Access programmes.
Phase 3 · Yr 3-4	ECOWAS Regional Rollout Target NAWEC (Gambia), SBEE (Benin), SENELEC (Senegal). Position GridIQ as the de facto ECOWAS utility intelligence standard. Monetise regional EIS data feed as standalone subscription product for ECOWAS Secretariat and DFIs.
Phase 4 · Yr 5+	Continental Platform Licensing License platform and AI models to utilities in East and Southern Africa (EAPP/SAPP corridors). Partner with major DFIs to embed GridIQ as a standard component of energy access financing covenants.

DFI as channel, not just funder: By building DFI-format reporting natively into the platform, GridIQ makes itself a procurement preference for any AfDB, IFC, or EU-funded utility modernisation programme. Each grant-funded deployment is also a sales case study for the next.

09 Technology Stack

GridIQ's architecture spans hardware edge, cloud data processing, AI/ML modelling, and standards-based API integration — designed from the ground up for the infrastructure constraints of West African utility environments.



10 Team & Organisation

GridIQ West Africa is a joint venture between three complementary organisations, combining deep West African utility relationships, energy sector technical expertise, and AI/software engineering capability.

CO-FOUNDER
RenewBiz

West Africa energy business development and utility partnership lead. Primary interface with ECOWAS member utilities and DFI programme teams.

CO-FOUNDER
AFCEN

African Centre for Energy and Natural resources. Technical advisory, regulatory engagement, and ECOWAS Secretariat relationships.

CO-FOUNDER
Smart Innovation

AI/ML development, IoT hardware integration, cloud platform engineering, and software product delivery.

Key functional roles required across the venture include: Platform Engineering Lead, AI/ML Data Scientist (NTL specialisation), IoT Hardware Engineer, GIS & Geospatial Analyst, Utility Partnerships Manager (Francophone West Africa), and DFI Grants & Reporting Specialist.

Organisational advantage: The three-partner structure deliberately combines what no single organisation in the market currently offers — deep utility-side relationships (RenewBiz), ECOWAS regulatory access (AFCEN), and the AI/IoT engineering capability to actually build and maintain the platform (Smart Innovation). This coalition is itself a competitive barrier.